

Copies, light and honest accuracy: auditing a public eyelid-anaemia dataset and building a calibrated phone check

PaleCheck project · code and data: github.com/d3osaju/palecheck

Abstract

Phone apps promise to detect anaemia from a photo of the inner eyelid, and the largest public dataset for the task, CP-AnemiC, reports up to AUC 0.835. We audited its 710 images. **571 (80%)** are copies of a photograph that also appears under another patient record: 312 pixel-identical, 12 re-encoded and 247 re-cropped or mirrored. The 710 records contain only **308 distinct photographs**. Under random cross-validation a 1-nearest-neighbour "memoriser" scores AUC 0.88 on the published data and 0.59 once all copies are removed, while a simple colour model stays near 0.68. On the 131 independent images, nested cross-validation gives AUC 0.65 (95% CI 0.54–0.73). A physically modelled warm bulb shifts eyelid redness 3.8× more than anaemia does; a printed white card removes the shift. We release PaleCheck, an offline, multilingual phone app built around these limits; the audit tools DupeScope and the Benchmark Lab; and cleaned dataset manifests.

1 Introduction

Anaemia affects 59.1% of Indian girls aged 15–19 [2]. Clinicians look for pallor of the palpebral conjunctiva (inner lower eyelid), and several studies train image models on conjunctiva photographs [1, 5]. Reported accuracies are high, but small medical datasets are prone to *leakage*, where test images are not independent of training images, a well-documented cause of inflated results across machine-learning-based science [7]. We set out to build a screening app and audited the data first.

2 Data and audit

CP-AnemiC contains 710 cropped conjunctiva images of children aged 6–59 months from 10 Ghanaian hospitals, each linked to a laboratory haemoglobin value; anaemia is $Hb < 11 \text{ g/dL}$ [1, 6]. We searched for photographs reused across records in three passes:

1. **Identical:** MD5 of decoded RGBA pixels. 312 images in 87 groups (up to 11 copies; Hb within one group spans up to 7.1 g/dL).
2. **Near-identical:** 64-bit dHash within 6 bits, aspect ratio within 5%, mean absolute pixel difference ≤ 6.0 on a 128x64 white-flattened grid. 12 more images.
3. **Re-cropped:** same photograph, different outline: at the best translation, RMS colour difference ≤ 4.0 over an overlap of $\geq 50\%$ of the smaller cut-out; candidates are each image's 8 nearest colour neighbours; mirrored orientation also tested. 247 more images in 135 clusters (3 mirror-image pairs).

As a control for pass 3 we re-scored the same candidate pairs with one image mirrored, which preserves colour and texture statistics but destroys a true pixel match. At the threshold the control matched only genuine mirror-image copies (difference 0); the next-closest control pair lay above the threshold. Every copy cluster lists different patients (hospital, age, sex or haemoglobin), so no record in a cluster can be verified; all 571 images were excluded.

Table 1. Copies of a photograph filed under more than one record.

Pass	Images added	Cumulative excluded	Images kept
Identical	312	312	398
+ Near-identical	12	324	386
+ Re-cropped / mirrored	247	571	139

3 Methods

Features. Tissue pixels (outline mask, glare and shadow removed) are converted sRGB $\rightarrow$ linear $\rightarrow$ XYZ $\rightarrow$ CIELAB (D65). Seventeen colour features are computed: absolute colour (mean and 25th-percentile a^* , L^* , b^* , erythema index, chromaticity, saturation, a^* spread) and within-image contrasts (percentile differences of a^* and erythema index, and others). **Model.** L2-regularised logistic regression fitted by Newton's method, λ by inner 5-fold CV [8]. **Protocol (v3).** After exclusion, 8 demo images are held out; the rest are split 70/30 into development and a lockbox, stratified by label. The feature set (absolute, relative or all) and λ are chosen by nested CV on development only; the lockbox is evaluated once. We also report nested 5 $\times$ 5-fold CV with 5 repeats over all 131 images, leave-one-hospital-out, subgroups and bootstrap 95% CIs (2,000 resamples). **Leakage demonstration.** At each cleaning level, a memoriser (1-nearest-neighbour on standardised absolute features) and the colour model ($\lambda = 1$) are compared under random 5-fold CV. **Lighting.** Eyelids are re-lit with blackbody illuminants (Planckian-locus fit [4]) by von Kries scaling in linear sRGB, and corrected by dividing by a white card photographed under the same light. **Software.** NumPy pipeline; a C# re-implementation in the app is unit-tested to reproduce the Python features to 10^{-9} ; CI re-downloads the dataset and re-derives every number.

4 Results

4.1 Copies inflate benchmark scores

Table 2. AUC under random 5-fold CV (5 repeats). The memoriser knows nothing about anaemia.

Data	Images	Memoriser	Colour model
published	710	0.88	0.55
identical removed	398	0.80	0.65
near-identical removed	386	0.79	0.65
re-crops removed	139	0.59	0.68

On the published data, 42% of test images have a pixel-identical twin in the training folds (identical-copy layer alone), and a twin-copying rule reaches 70.7% accuracy. The memoriser's advantage persists after removing identical and near-identical copies and disappears only when re-crops are removed, while the colour model is stable across all levels, consistent with copies, not colour, driving the memoriser.

4.2 Honest accuracy

Table 3. Results by protocol version. Each version re-ran the full protocol with a new random lockbox.

Version	Images kept	Nested-CV AUC	Lockbox AUC (95% CI)
v1: exact duplicates removed	398	0.64	0.68 (0.58–0.78, n = 117)
v2: identical and near-identical copies removed	386	0.65	0.65 (0.55–0.75, n = 113)
v3: identical, near-identical and re-cropped copies removed	139	0.65	0.73 (0.56–0.89, n = 39)

Nested CV is stable at about 0.65 across versions (v3 pooled 95% CI 0.54–0.73); the v3 lockbox (0.73, n = 39) is too small to narrow this. Leave-one-hospital-out AUC is 0.61. Development data selected the "relative" feature set (absolute 0.63, relative 0.65, all 0.59). Estimating the haemoglobin value itself did not beat predicting the mean (MAE 1.63 vs 1.70 g/dL).

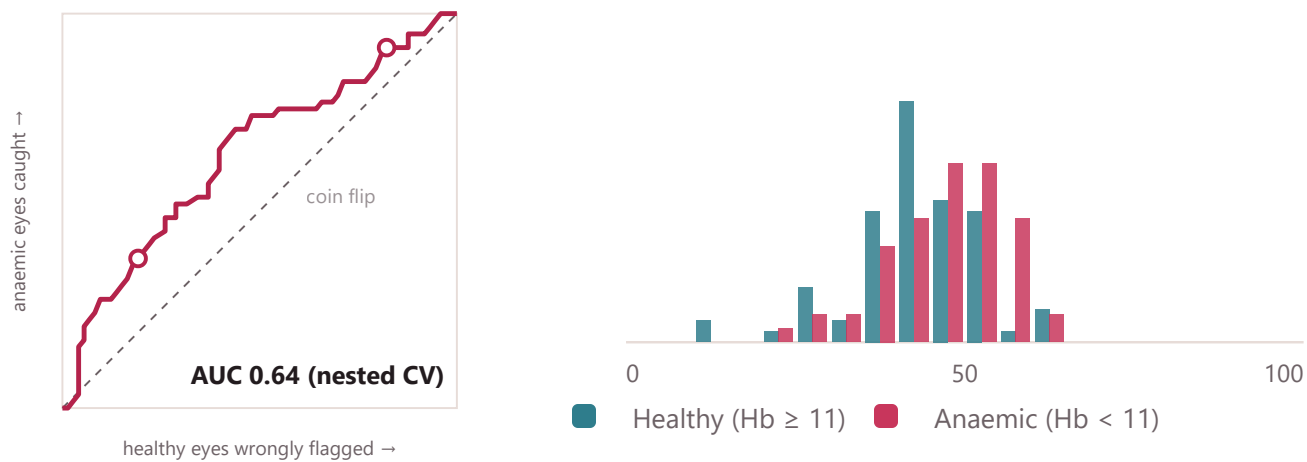


Figure 1. Left: ROC curve (nested CV), with the app's two cut-offs. Right: out-of-fold pallor index of healthy and anaemic reference eyes.

4.3 Lighting

Table 4. Mean shift in eyelid redness (a^*) when re-lit, relative to the healthy-vs-anaemic gap of 3.9.

Light	a^* shift	$\times$ anaemia gap	After card correction
2700 K	+14.7	3.8 $\times$	0.0
3500 K	+9.6	2.4 $\times$	0.0
4000 K	+7.3	1.9 $\times$	0.0
5000 K	+4.1	1.1 $\times$	0.0

In simulation the card correction is exact by construction; a real camera adds auto white balance and tone curves. The app's Lighting Lab measures the residual on a real phone (protocol in the repository).

4.4 Should a photo decide who is tested?

At the app's cautious cut-off (pallor index ≥ 35 ; sensitivity 91%, specificity 14%) and 59.1% prevalence, photo-first screening of 1,000 girls needs 893 blood tests instead of 1,000 but misses 51 of 591 anaemic girls. Where anaemia is this common, a photo should not replace testing; PaleCheck is positioned as a prompt to get tested for people outside school screening.

5 The PaleCheck app

A Blazor WebAssembly progressive web app (English, Malayalam, Hindi) that runs offline; photos never leave the device. It detects a printed white calibration card automatically, selects the eyelid with one tap (region growing in CIELAB), rejects glare and shadow, corrects for the room light, averages up to five photos and reports a pallor index with an explicit overlap zone, never a diagnosis. Every result links to the free Anaemia Mukh Bharat haemoglobin test [3]; a symptom checklist recommends testing regardless of the photo. Two research tools ship with it: DupeScope (in-browser duplicate and re-encoding detection for any image dataset) and the Benchmark Lab (Table 2, reproducible live).

6 Limitations

- Only 139 independent images remain; all estimates have wide intervals.
- The re-crop search compares each image with its 8 nearest colour neighbours, so some copies may remain.
- Reference eyes are young children in Ghana photographed without a colour reference; Indian teenagers need a new, calibrated dataset and the 12 g/dL threshold [6].
- Lighting results are simulated; real-camera validation is pending. Not a medical device.

7 Ethics, data and AI use

No new human data were collected. CP-AnemiC is used under CC BY 4.0; we publish only derived manifests and colour features. The app stores numbers, never photos, and only on the user's device. Code and text were developed with the help of an AI coding assistant; every number in this report is generated by the published scripts and re-checked by continuous integration. We intend to share the audit with the dataset's authors.

References

1. Appiahene P, Chaturvedi K, Asare JW, Donkoh ET, Prasad M. CP-AnemiC: a conjunctival pallor dataset and benchmark for anemia detection in children. *Medicine in Novel Technology and Devices*, 2023. doi:10.1016/j.medntd.2023.100244. Data: doi:10.17632/m53vz6b7fx.1.
2. Scott S, et al. Anaemia in Indians aged 10–19 years: prevalence, burden and associated factors at national and regional levels. *Maternal & Child Nutrition*, 2022. doi:10.1111/mcn.13391.
3. Ministry of Health and Family Welfare, Government of India. Anemia Mukh Bharat: operational guidelines. National Health Mission, 2018.
4. Kang B, et al. Design of advanced color temperature control system for HDTV applications. *Journal of the Korean Physical Society*, 2002.
5. Dimauro G, et al. An intelligent non-invasive system for automated diagnosis of anemia exploiting a novel dataset. *Artificial Intelligence in Medicine*, 2023.

6. World Health Organization. Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity. WHO/NMH/NHD/MNM/11.1, 2011.
7. Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 2023.
8. Varma S, Simon R. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 2006.